

Generating Tsunami Energy-Dissipating Barriers Using Machine Learning

1 Project Description

This project will develop a computational workflow for the design and assessment of tsunami energy-dissipating barrier geometries. The study is structured as a focused ten-week proof-of-concept investigation built around a single benchmark tsunami scenario, a defined optimisation framework, and a final structural verification stage. This scope is intended to ensure that the project remains technically ambitious while still achievable within the duration of the studentship.

1.1 Phase I: Benchmark Tsunami Setup and Data Acquisition

The first phase will establish the hydrodynamic modelling framework in MATLAB. A benchmark tsunami case based on earthquake forcing will be defined, with the 2011 Tohoku event providing the reference basis for the initial condition and bathymetric context. The simulation will model tsunami propagation, nearshore shoaling, inundation, and interaction with a reference geometry introduced for data extraction purposes. This stage will generate the hydrodynamic quantities required for optimisation, including free-surface elevation, flow velocity, momentum flux, and other performance indicators relevant to tsunami loading and energy dissipation.

1.2 Phase II: Machine-Learning-Guided Geometry Optimisation

The second phase will form the core iterative design loop of the project. A machine-learning-guided optimisation procedure will generate candidate barrier geometries subject to predefined hydraulic, geometric, and practical constraints. For each candidate design, the tsunami interaction will be resimulated and the resulting hydrodynamic performance metrics will be extracted. These results will then be used to refine the geometry in subsequent iterations. The objective of this phase is to identify barrier forms that improve tsunami energy dissipation, reduce transmitted force and pressure, and weaken damaging flow structures. The optimisation will continue until prescribed convergence criteria are met, thereby producing a shortlisted optimum geometry for further assessment.

1.3 Phase III: Structural Verification and Final Validation

The final phase will assess how the forces and loads from the tsunami are distributed and dispersed on the structure. The shortlisted optimum design will therefore be transferred into ANSYS or Abaqus, where the tsunami-induced loads obtained from the fluid simulations will be applied to evaluate stress distributions, deformation patterns, and structural energy absorption. This stage will function as a targeted verification exercise, allowing the project to distinguish between designs that perform well from a hydrodynamic perspective and those that remain viable from a structural engineering standpoint.

2 Expected Contribution

The expected outcome of the project is a feasible multistage workflow for tsunami barrier design that integrates benchmark tsunami simulation, machine-learning-guided geometry optimisation, and structural verification. Within the ten-week timeframe, the project is expected to deliver a validated benchmark simulation, an iterative optimisation framework for candidate barrier generation, and an initial finite element assessment of the final shortlisted design. These outputs will provide a credible proof of concept for the use of computational optimisation in the development of next-generation tsunami damping structures.

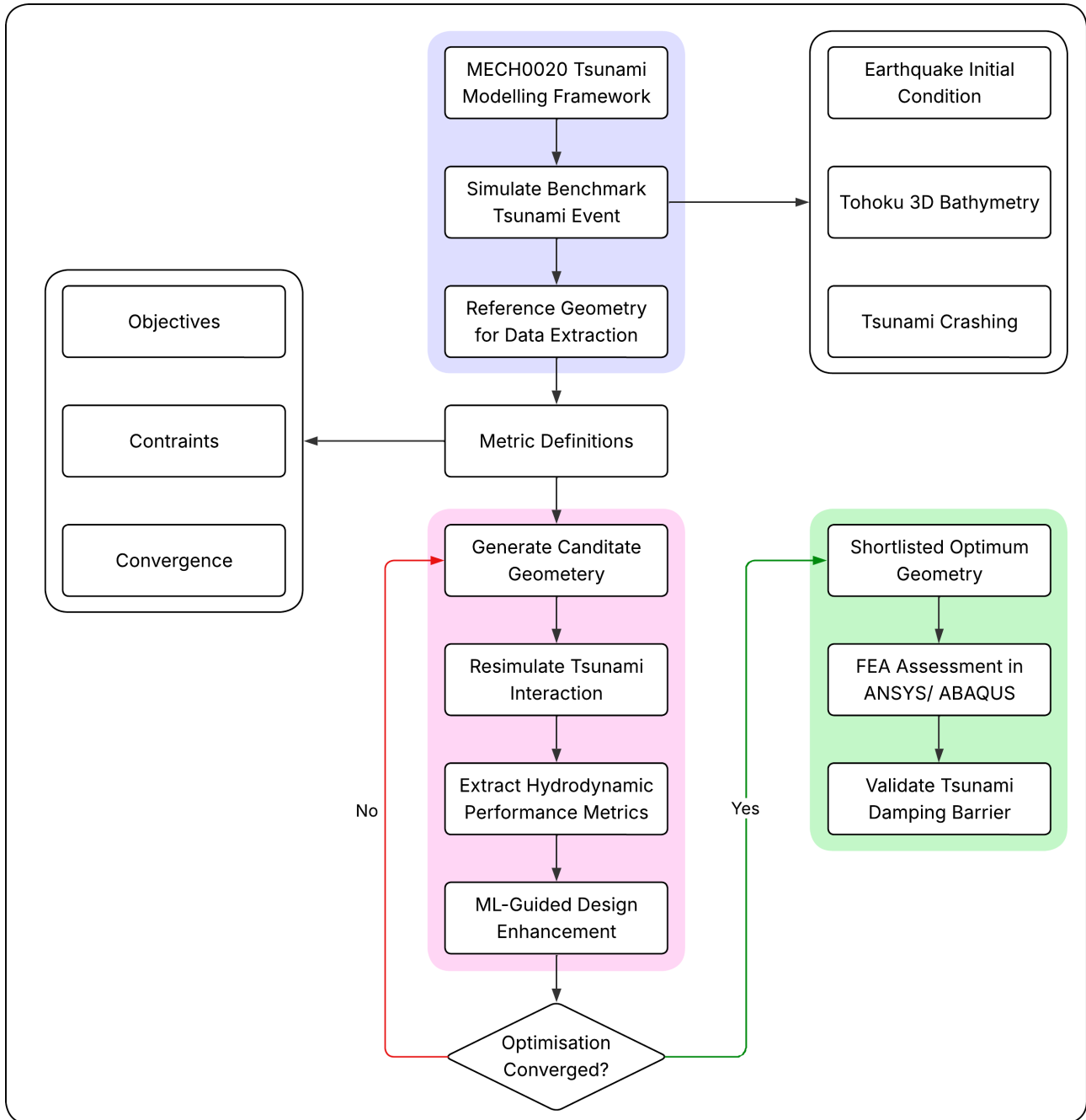


Figure 1: Flow chart of the proposed tsunami barrier optimisation framework, with the three principal phases highlighted. Phase I, *Benchmark Tsunami Setup and Data Acquisition*. Phase II, *Machine-Learning-Guided Geometry Optimisation*. Phase III, *Structural Verification and Final Validation*.